THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · EASY LEVEL CHAPTER TEST SET 2

Chapter 8: **Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Minimum time gap for an echo is: 1

(a) 0.01 s **(b)** 0.1 s **(c)** 1 s **(d)** 10 s

2. f = 50 Hz, lambda = 2 m. Speed: 1

(a) 52 m/s **(b)** 100 m/s **(c)** 25 m/s **(d)** 48 m/s

3. SI unit of frequency: 1

(a) newton **(b)** hertz **(c)** metre **(d)** second

4. Bats use: 1

(a) smell **(b)** light **(c)** ultrasound **(d)** infrasound

5. v = 340 m/s, f = 170 Hz. Wavelength: 1

(a) 170 m **(b)** 340 m **(c)** 0.5 m **(d)** 2 m

6. Pitch depends on: 1

(a) colour **(b)** frequency **(c)** speed **(d)** amplitude

7. SONAR uses: 1

(a) radio waves **(b)** light **(c)** infrasound **(d)** ultrasonic waves

8. v = 340 m/s, lambda = 1 m. Frequency: 1

(a) 34 Hz **(b)** 0.003 Hz **(c)** 340 Hz **(d)** 1 Hz

9. Loudness depends on: 1

(a) amplitude **(b)** wavelength **(c)** speed **(d)** frequency

10. Thunder is heard after lightning because: 1

(a) light is faster than sound **(b)** lightning is far
(c) both are same **(d)** sound is faster

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is an echo? 2

12. State the audible range. 2

13. What is frequency? 2

14. What is pitch? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. State three characteristics of sound. 3

16. Name three applications of ultrasound. 3

17. Which part of a bell vibrates to produce sound? Give three more examples of vibrating parts. 3

OR

Find the speed: f = 100 Hz, lambda = 3 m. 3

18. State three ways to reduce noise. 3

OR

State three things which sound waves carry. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain pitch, loudness and quality. 5

OR

Explain reverberation and how to reduce it. 5

20. Explain echo with conditions and an example. 5

OR

Describe an activity to show sound is produced by vibration. 5

END OF QUESTION PAPER — Total: 40 Marks

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